

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Electrical and Computer Engineering SEMESTER - 7 Winter 2025 (Regular)

Course : Bachelor of Technology (Electrical and Computer Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 7

Subject Code & Name: BTECC701 - ELECTRICAL DRIVES

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 The main function of a power modulator in an electrical drive is to
 - A. Sense motor speed and current
 - B. Convert and regulate power supplied to the motor as per load requirement
 - C. Provide mechanical coupling between motor and load
 - D. Store electrical energy for transient operation
 - 2 In the fundamental torque equation of an electrical drive with constant inertia, the developed motor torque is used to
 - A. Overcome friction torque only
 - B. Supply load torque only
 - C. Supply load torque and dynamic torque during transients
 - D. Compensate windage torque only
 - 3 In four-quadrant operation of an electrical drive, forward braking corresponds to
 - A. Positive speed and positive power
 - B. Positive speed and negative power
 - C. Negative speed and positive power
 - D. Negative speed and negative power
 - 4 In short time duty of an electrical drive, the motor is
 - A. Operated continuously until thermal steady state is reached
 - B. Operated for a duration shorter than the heating time constant and allowed to cool to ambient temperature before the next operation
 - C. Operated with periodic speed changes without rest
 - D. Operated with continuous loading and electrical braking
 - 5 In an electrical drive employing a closed-loop current (torque) control, the main reason for using an inner current control loop is to
 - A. Improve steady-state speed regulation
 - B. Protect the motor from excessive voltage
 - C. Achieve fast torque response and limit peak motor current
 - D. Eliminate the effect of load torque variations

- 6 In an electrical drive system, the equilibrium operating point is achieved when
 - A. Motor speed is maximum
 - B. Motor power is zero
 - C. Motor torque equals load torque
 - D. Load torque exceeds motor torque
- 7 In a single-phase full-wave controlled rectifier-fed DC motor drive, the frequency of torque fluctuations is
 - A. Equal to the supply frequency
 - B. Twice the supply frequency
 - C. Half of the supply frequency
 - D. Independent of the supply frequency
- 8 In a chopper-fed DC motor drive, the motor speed is controlled mainly by
 - A. Varying the supply voltage
 - B. Changing the armature resistance
 - C. Varying the duty ratio of the chopper
 - D. Changing the motor field flux only
- 9 In constant V/f control of an induction motor, maintaining the ratio of supply voltage to frequency constant primarily ensures
 - A. Constant slip over the entire speed range
 - B. Constant air-gap flux and approximately constant torque capability
 - C. Reduction in stator copper losses
 - D. Elimination of harmonic currents
- 10 In a slip power recovery scheme used with a wound-rotor induction motor, the main advantage is
 - A. Improvement in starting torque without external resistance
 - B. Reduction of rotor copper losses by feeding slip power back to the supply
 - C. Achievement of super-synchronous speed operation
 - D. Elimination of reactive power consumption
- 11 In a load-commutated converter (LCC) fed synchronous motor drive, commutation of thyristors is mainly achieved by
 - A. External forced commutation circuits
 - B. Supply voltage reversal
 - C. Motor-induced back EMF
 - D. DC link current chopping
- 12 In a synchronous motor drive, the distinguishing feature between true synchronous control and self-control mode is that in self-control mode
 - A. The motor speed is independent of supply frequency
 - B. The inverter frequency is synchronized with rotor position
 - C. The motor always operates at constant slip
 - D. The field current is not required

Q2. Solve the following.

- A) Discuss the working of power modulator. Explain in detail different types of power modulators. 6
- B) Explain various components of load torque. 6

Q3. Solve the following.

- A) Explain the operation of a closed loop torque control and current limit control scheme with neat diagram. 6
- B) Explain different classes of duty depending on loads. 6

- Q4. Solve Any Two of the following.
- A) Explain the operation of single phase fully controlled rectifier control of DC Separately Excited motor operating in continuous mode of conduction. Derive speed-torque relation. 6
 - B) Explain chopper control of separately excited dc motor during motoring operation. Derive the relation between voltage & duty cycle and motor speed & torque. Draw speed torque curve. 6
 - C) A 200V, 875rpm, 150A separately excited dc motor has an armature resistance of 0.06ohm. It is fed from a single phase fully controlled rectifier with an ac source of 220V, 50Hz. Assuming continuous conduction, calculate (i) Firing angle for rated motor torque and 750rpm. (ii) Motor speed for $\alpha=160^\circ$ and rated torque. 6
- Q5. Solve Any Two of the following.
- A) Explain V/f control method of IM drives. Draw corresponding speed torque characteristics. 6
 - B) The parameters of a three phase 400 Volts, 50 Hz, 6 pole, 960 rpm, and star connected induction motor has the following parameters per phase referred to the stator. $R_1 = 0.4\Omega$, $R_2 = 0.20\Omega$, $X_1 = X_2 = 1.5\Omega$, $X_m = 30\Omega$. If the motor is controlled by variable frequency control at a constant flux of rated value, determine the motor speed and the stator current at half the rated torque and 25Hz 6
 - C) Explain the concept of slip power recovery. Describe Static Kramer Drive. 6
- Q6. Solve Any Two of the following.
- A) Describe the open-loop and closed loop methods of speed control of a synchronous motor using VSI. 6
 - B) Explain Closed loop speed control of load commutated inverter fed synchronous motor drive. 6
 - C) State and explain the modes of speed control of synchronous motor on the basis of frequency. 6

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